

```
</> activity_main.xml MainActivity.kt x
1 package com.example.bharani
2
3 import android.os.Bundle
4 import androidx.appcompat.app.AppCompatActivity
5
6 </> class MainActivity : AppCompatActivity() {
7 @   override fun onCreate(savedInstanceState: Bundle?) {
8     super.onCreate(savedInstanceState)
9     setContentView(R.layout.activity_main)
10
11 }
```

× Create a project

Let's start with a name for your project[®]

Project name

☐ Join the [Google Developer Program](#) to enrich your developer journey with access to AI assistance, learning resources, profile badges, and more!

Already have a Google Cloud project?
[Add Firebase to Google Cloud project](#)

Continue

☐ Enable Google Analytics for this project
Recommended

Previous

Create project

bharani

Spark plan



Getting started? Tell Gemini

Get started by adding Firebase to your app

Android



Add an app to get started

1

②

```
com.example.bharani
```

②

My Android App

②

00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:(

❶ Required for Dynamic Links, and Google Sign-In or phone number support in Auth.
Edit SHA-1s in Settings.

Register app

Android package name: com.example.bharani

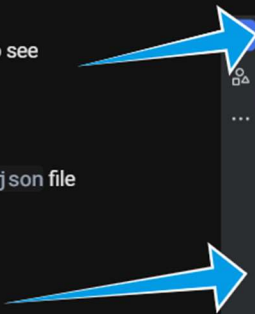
2

[Unity](#) [C++](#)

[Download google-services.json](#)

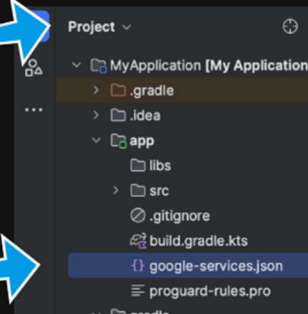
Switch to the **Project** view in Android Studio to see your project root directory.

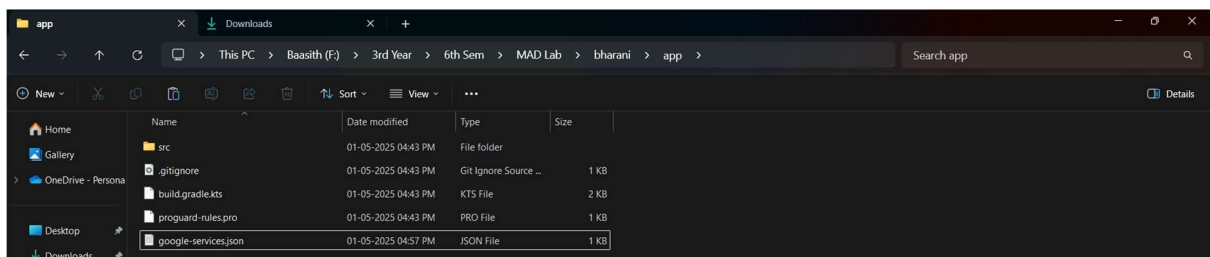
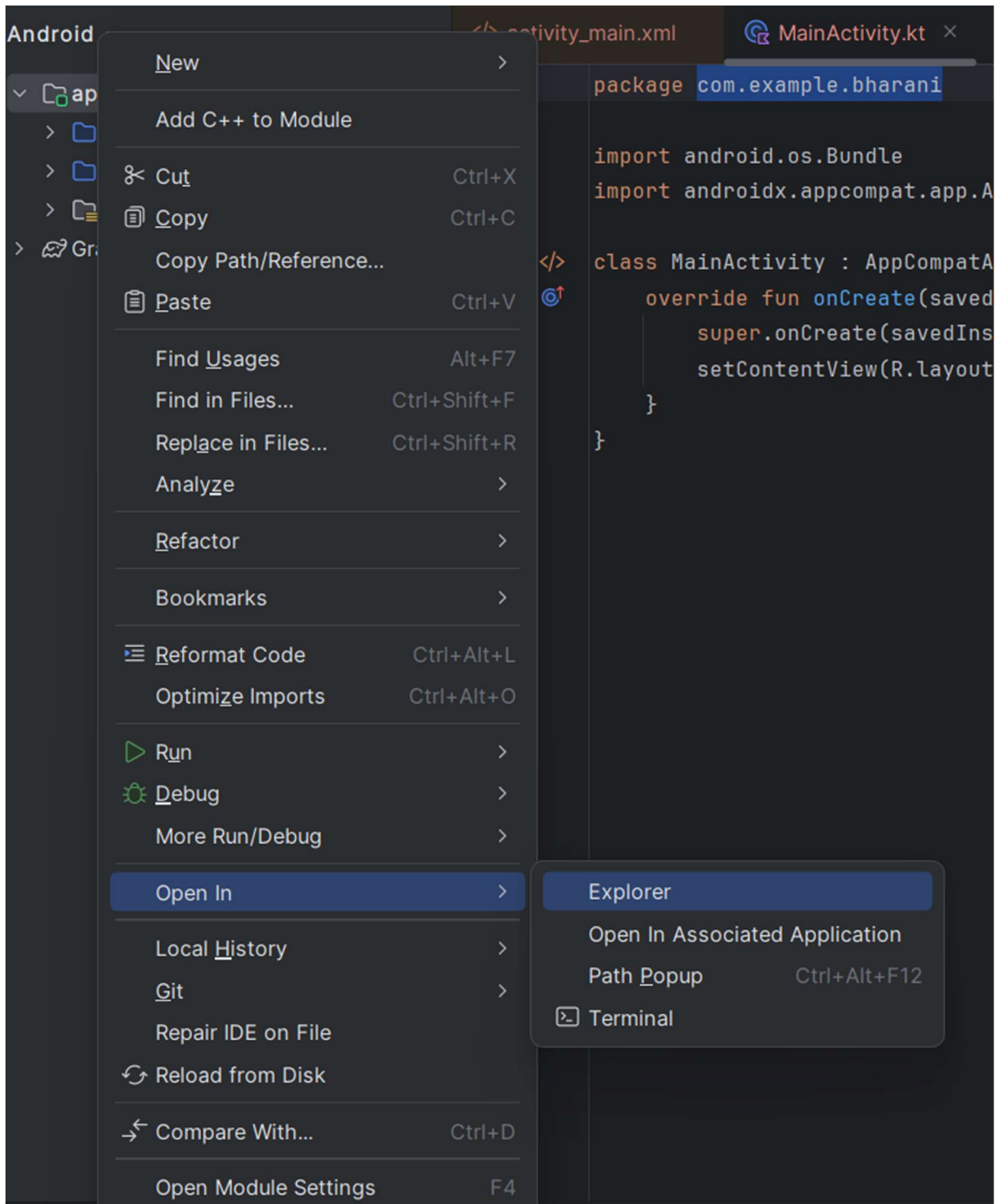
Move your downloaded `google-services.json` file into your module (app-level) root directory.



google-services.json

Next





```

plugins {
    // ...

    // Add the dependency for the Google services Gradle plugin
    id("com.google.gms.google-services") version "4.4.2" apply false
}

```

Then, in your module (app-level) `build.gradle.kts` file, add both the `google-services` plugin and any Firebase SDKs that you want to use in your app:

Module (app-level) Gradle file (`<project>/<app-module>/build.gradle.kts`):

```

plugins {
    id("com.android.application")
    // Add the Google services Gradle plugin
    id("com.google.gms.google-services")
    ...
}

dependencies {
    // Import the Firebase BoM
    implementation(platform("com.google.firebase:firebase-bom:33.13.0"))

    // TODO: Add the dependencies for Firebase products you want to use
    // When using the BoM, don't specify versions in Firebase dependencies
    // https://firebase.google.com/docs/android/setup#available-libraries
}

```

Gradle files have changed since last project sync. A project sync may be necessary for the IDE to work properly.

```

1 // Top-level build file where you can add configuration options common to all sub-projects/modules.
2 plugins {
3     alias(libs.plugins.android.application) apply false
4     alias(libs.plugins.kotlin.android) apply false
5     id("com.google.gms.google-services") version "4.4.2" apply false
6 }

```

Gradle files have changed since last project sync. A project sync may be necessary for the IDE to work properly. Sync Now Ignore these changes

```

1 plugins {
2     alias(libs.plugins.android.application)
3     alias(libs.plugins.kotlin.android)
4     id("com.google.gms.google-services")
5 }
6

```

```
</> activity_main.xml MainActivity.kt build.gradle.kts (bharani) build.gradle.kts (:app) ×
Gradle files have changed since last project sync. A project sync may be necessary for the IDE to work properly.

7  android {
21      buildTypes {
22          release {
23              // TODO: Add your own build configurations here.
24          }
25      }
26      compileOptions {
27          sourceCompatibility = JavaVersion.VERSION_11
28          targetCompatibility = JavaVersion.VERSION_11
29      }
30      kotlinOptions {
31          jvmTarget = "11"
32      }
33  }
34
35  dependencies {
36      implementation(libs.androidx.core.ktx)
37      implementation(libs.androidx.appcompat)
38      implementation(libs.material)
39      implementation(libs.androidx.activity)
40      implementation(libs.androidx.constraintlayout)
41      testImplementation(libs.junit)
42      androidTestImplementation(libs.androidx.junit)
43      androidTestImplementation(libs.androidx.espresso.core)
44      implementation(platform("com.google.firebase:firebase-bom:33.13.0"))
45  }
```

```
package com.example.bharani

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val db = Firebase.firestore

    }
}
```



```
package com.example.bharani

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val db = Firebase.firestore
    }
}
```

Unresolved reference: Firebase

[Add library dependency and import](#) Alt+Shift+Enter [More actions...](#) Alt+Enter

Suggested Imports

`com.google.firebase:firebase-common-ktx`

`com.google.firebase:firebase-common-ktx`

```
val db = Firebase.firestore
```

Unresolved reference: firestore

[Add dependency on com.google.firebase:firebase-firestore-ktx and import](#) Alt+Shift+Enter [More actions...](#) Alt+Enter



Firebase



Project Overview



Product categories

Build



App Check



App Hosting



Authentication



Data Connect



Extensions



Firestore Database



Functions



Hosting



Machine Learning



Realtime Database



Storage

Cloud Firestore

Realtime updates, powerful queries, and automatic scaling

Create database

 Ask Gemini

Create database



1 Set name and location — 2 Secure rules



Interested in Firestore with MongoDB compatibility? Create a Firestore Enterprise database in the Google Cloud Console

[Learn more](#)

Database ID

(default)

Location

nam5 (United States)

 Your location setting is where your Cloud Firestore data will be stored



After you set this location, you cannot change it later.

[Learn more](#)

Cancel

Next

Create database



- 1 Set name and location — 2 Secure rules

After you define your data structure, you will need to write rules to secure your data.

[Learn more](#)

☐ Start in production mode

Your data is private by default. Client read/write access will only be granted as specified by your security rules.

☒ Start in test mode

Your data is open by default to enable quick setup. However, you must update your security rules within 30 days to enable long-term client read/write access.

```
rules_version = '2';

service cloud.firestore {
  match /databases/{database}/documents {
    match /{document=**} {
      allow read, write: if
        request.time < timestamp.date(2025, 5, 31);
    }
  }
}
```

! The default security rules for test mode allow anyone with your database reference to view, edit and delete all data in your database for the next 30 days

Cancel

Create

Start a collection

- 1 Give the collection an ID — 2 Add its first document

Parent path

/

Collection ID

bharani

Cancel

Next

Start a collection

✓ Give the collection an ID

2 Add its first document

Document parent path

/bharani

Document ID ?

cart

⊕ Add field

Cancel

Save

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
```

```
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.696"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.186" />
```

<EditText

```
android:id="@+id/age"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="text"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.701"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.338" />
```

<EditText

```
android:id="@+id/contact"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="text"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.706"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
```

```
app:layout_constraintVertical_bias="0.485" />
```

```
<Button
```

```
    android:id="@+id/submit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Submit"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.154"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.622" />
```

```
<Button
```

```
    android:id="@+id/retrieve"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Retrieve Data"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.735"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.622" />
```

```
<TextView
```

```
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Name"
```

```
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.107"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.195" />
```

```
<TextView
```

```
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Age"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.107"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.341" />
```

```
<TextView
```

```
    android:id="@+id/textView3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Contact"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.107"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.478" />
```

```
<TextView
    android:id="@+id/result"
    android:layout_width="393dp"
    android:layout_height="207dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.944" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.kt

```
package com.example.bharani
```

```
import android.os.Bundle
import android.util.Log
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import com.google.firebase.Firebase
import com.google.firebase.firestore.FieldValue
import com.google.firebase.firestore.firestore
```

```
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
```



```

val itemNameEditText = findViewById<EditText>(R.id.name)
val age = findViewById<EditText>(R.id.age)
val contact = findViewById<EditText>(R.id.contact)
val addButton = findViewById<Button>(R.id.submit)
val result = findViewById<TextView>(R.id.result)
val retrieveButton = findViewById<Button>(R.id.retrieve)

val db = Firebase.firestore

addButton.setOnClickListener {
    val itemName = itemNameEditText.text.toString().trim()
    val itemAge = age.text.toString().trim()
    val itemContact = contact.text.toString().trim()
    if (itemName.isNotEmpty()) {
        val userDocumentRef = db.collection("bharani").document("cart")

        userDocumentRef.update("items", FieldValue.arrayUnion(mapOf("name" to itemName,
"age" to itemAge, "contact" to itemContact)))

        .addOnSuccessListener {
            Toast.makeText(this, "Item added to cart", Toast.LENGTH_SHORT).show()
        }
        .addOnFailureListener { e ->
            Log.e("MainActivity", "Error adding item to cart", e)
            Toast.makeText(this, "Failed to add item to cart", Toast.LENGTH_SHORT).show()
        }
        itemNameEditText.text.clear()
        age.text.clear()
        contact.text.clear()
    } else {
        Toast.makeText(this, "Item name cannot be empty", Toast.LENGTH_SHORT).show()
    }
}

```

```
}
```

```
retrieveButton.setOnClickListener {
```

```
    db.collection("bharani").document("cart").get()
```

```
    .addOnSuccessListener { document ->
```

```
        if (document.exists()) {
```

```
            val items = document.get("items") as List<Map<String, Any>>
```

```
            for (item in items) {
```

```
                val itemName = item["name"] as? String
```

```
                val itemAge = item["age"] as? String
```

```
                val itemContact = item["contact"] as? String
```

```
                result.append("Name: $itemName, Age: $itemAge, Contact: $itemContact\n")
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
}
```

```
}
```